half-baked calculations, misty metaphysical discussions and equations that work well, but inexplicably, into a perfect architecture: airy, simple and extremely beautiful. Beautiful, but stratospherically abstract.

The venerable Bohr said of him, ‘Of all physicists, Dirac has the purest soul.’ And don’t his eyes, in figure 4.4 show so? His physics has the pristine clarity of a song. For him, the world is not made of things, it’s constituted of an abstract mathematical structure which shows us how things appear and how they behave when manifesting themselves. It’s a magical encounter between logic and intuition. Deeply impressed, Einstein remarked, ‘Dirac poses problems for me. To maintain an equilibrium along this vertiginous course, between genius and madness, is a daunting enterprise.’

Dirac’s quantum mechanics is the mathematical theory used today by any engineer, chemist or molecular biologist. In it, every object is defined by an abstract space[fn20] and has no property in itself, apart from those that are unchanging, such as mass. Its position and velocity, its angular momentum and its electrical potential, and so on, acquire reality only when it collides – ‘interacts’– with another object. It is not just its position which is undefined, as Heisenberg had recognized: *no* variable of the object is defined between one interaction and the next. The *relational* aspect of the theory becomes universal.

When it suddenly appears, in the course of an interaction with another object, a physical variable (velocity, energy, momentum, angular momentum) does not assume just any value. Dirac provides the general recipe to compute the set of values that a physical variable can take.[fn21] These values are analogous to the spectra of the light emitted by atoms. Today we call the set of the particular values which a variable may assume the ‘spectrum’ of that variable, by analogy with the spectra into which the light of elements decomposes – the first manifestation of this phenomenon. For example, the radius of the orbitals of an electron around a nucleus can acquire only specific values, those that Bohr had hypothesized, which form the ‘spectrum of the radius’.

The theory also gives information on *which* value of the spectrum will manifest itself in the next interaction, but only in the form of probabilities. We do not know with certainty where the electron will appear, but we can compute the *probability* that it will appear here or there. This is a radical change from Newton’s theory, where it is possible, in principle, to predict the future with certainty. Quantum mechanics brings probability to the heart of the evolution of things. This *indeterminacy* is the third cornerstone of quantum mechanics: the discovery that chance operates at the atomic level. While Newton’s physics allows for the prediction of the future with exactitude, if we have sufficient information about the initial data and if we